# MCQ Midterm BIO 202

Q
Glucose residues in amylose are linked by _____.
α (1,4) linkage β (1,4)
linkage
α (1,6) linkage
None of these

Q
The number of carbon atoms in lysine is _____.
4
6
8
10

Q
If the carbonyl group is present at the end of the monosaccharide then it is called _____.
Acid anhydride
Aldose
Ketose
None of these

Q
The number of stereoisomers for a molecule containing only one chiral carbon is _____.
16
2
4
8

Q
Seminal fluid is rich in _____.
Fructose
Lactose

Xylose
Arabinose

Q
Which of the following glycosidic linkage exists between two glucose units of maltose?

alpha 1,4
beta 1,4
alpha 1,6
alpha1,4 and alpha 1,6

Q
Amino sugar is formed by the replacement of the hydroxyl group at ------ of the parent monosaccharide with amino group.
C-1

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C-2
C-3
C-6

Q
Which one of the following is a structural homopolysaccharide?
Cellulose
Starch
Glycogen
All of these
Q
Which one of the following amino acids is not specified by three letter codon?
Serine
Proline
Selenocysteine
Methionine

Q:
Acid dissociation constants are designated as _____.

PKa
K
Pa
None of these

Q
The pI of glutamate is _____.

1.2
5.6
3.22

6.7

Q
Which of the following amino acid is negatively charged at physiological pH?
Aspartate
Lysine
Alanine
Proline

Q
Which of the following refers to particularly stable arrangements of amino acid residues giving rise to recurring structural patterns?
Primary structure
Secondary structure
Tertiary structure
Quaternary structure

Q
Keratin is present in _____.

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Hair
Nail
Hoof
All of these

Q
Which of the following amino acid is involved in the hydrophobic interactions in the tertiary structure of protein?
Phenylalanine
Cysteine
Glycine
None of these

Q
The free energy of a protein molecule is influenced by _____.
Hydrophobicity
hydrogen bonds
electrostatic interactions
All of these

Q
Which of the following diseases is caused by protein misfolding?
Alzheimer's disease
Huntington's disease
Parkinson's disease
All of these

Q
Iron is a _____.

non-metal
metalloid
transition metal
halogen

Q
Myoglobin can bind _____ molecules of oxygen.

One
Two
Three
Four
Q
The partial pressure of oxygen (pO2) in lungs is about _____.
30 mmHg
50 mmHg
100 mmHg
200 mmHg

Q
In sickle cell anemia, Glutamate is replaced with _____.

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Arginine
Alanine
Methionine
Valine

Q
Oxy hemoglobin dissociation curve describe the relationship of ----------
Available Oxygen
Amount of carbon dioxide carried by hemoglobin
Fibrinogen
Oxy-hemoglobin curve stability
Q
Which one of the following is made up of two molecules of glucose?
Maltose
Starch
Cellulose
Amylose

Q
Which bond exists between the carbonyl group and oxygen of a hydroxyl group in ring structure of glucose?

Ionic
Covalent
hydrophobic interaction
both ionic and covalent

Q
In Benedict test, the formation of cuprous oxide is indicated by the formation of _____.
blue ppt
silver mirror
red ppt
none of these

Q
Glucose 6 Phosphate is formed by the addition of  phosphate group to C6 of glucose by _____ linkage.
Ether
Ester
Ionic
none of these

Q
E. coli convert lactose of milk to lactic acid by action of _____.
Beta galactosidase
Amyloglucosidase
Alpha amylase
Maltase

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Q
At any pH above pI, gycine has a net _____ charge and will move towards the _____.
negative, anode
negative, cathode
positive, anode
positive, cathode

Q
Which of the following amino acids contain two COOH groups?
Alanine
Glutamate
Glycine
Valine

Q
Which of the following amino acid is positively charged at physiological pH?
Lysine
Aspartate
Tyrosine

Alanine

Q
Peptide bond is _____ linkage between two amino acids.
Ester
Amide
Ether
None of these

Q
Which of the following shows the greatest tendency to form α helices in most experimental model systems?
Glycine
Alanine
Proline
Serine

Q
Immunoglobulin G (IgG) consists of _____.
Two heavy chains and two light chains
Four heavy chains
Four light chains
None of these

Q
The free energy of a protein molecule is influenced by _____.
Hydrophobicity
hydrogen bonds
electrostatic interactions
All of these

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Q
Proteins are denatured by _____.
Temperature
Ph
organic solvents
temperature, pH, organic solvents

Q
About _____ of amino acid residues in the myoglobin are found in alpha helices.
98%
78%
27%
Q
Hemoglobin is found exclusively in _____.
white blood cells
red blood cells
platelets
platelets and white blood cells

Q
The partial pressure of oxygen (pO2) in lungs is about _____.
30 mmHg
50 mmHg
100 mmHg
200 mmHg

Q
In sickle cell anemia, Glutamate is replaced with _____.
Alanine
Methionine
Valine

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